

Het Patel

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Summary

MS in Computer Science student with hands-on experience in **machine learning**, **generative AI**, and **software engineering**. Built **Python/PyTorch ML pipelines**, transformer-based models, and full-stack applications using React, Node.js, Express, and MongoDB. Experienced in **data preprocessing**, **model training/evaluation**, classification metrics, experiment tracking, debugging, and presenting technical findings for real-world product and research use cases.

Education

Illinois Institute of Technology

M.S. in Computer Science

Chicago, IL

Expected Dec 2026

Sardar Vallabhbhai Institute of Technology

B.S. in Computer Science

Anand, India

May 2024

Experience

Research Assistant, Illinois Institute of Technology

Jan 2026 – Present

- Built and evaluated **Python/PyTorch machine learning pipelines** across **5+ prototype variants**, comparing model quality, runtime behavior, and forecasting metrics to support model-selection decisions.
- Automated **data preprocessing**, experiment execution, validation, and result aggregation using Python scripts, improving reproducibility across repeated training and evaluation runs.
- Evaluated transformer/LLM-based model outputs, identified preprocessing and configuration issues, and documented error patterns to improve follow-up experimentation.
- Organized experiment results, charts, and performance notes into structured summaries for review, research discussion, and technical presentation.

Teaching Assistant, Illinois Institute of Technology

Sep 2025 – Jan 2026

- Supported 40+ students through labs, grading, and office hours, applying consistent evaluation criteria and giving clear, structured feedback on technical work.
- Reviewed programming assignments for correctness, debugging approach, documentation quality, and problem-solving clarity.

Software Engineer / Machine Learning Engineer, Barodaweb

Jun 2024 – Dec 2024

- Contributed to a **camera-integrated digital signage platform** using computer vision and image-processing logic to estimate viewer attributes and trigger personalized content in near real time.
- Built and tested **full-stack web application components** using React, Node.js, Express, and MongoDB for data capture, configuration workflows, and operational dashboards.
- Reviewed system behavior under different lighting, positioning, and deployment conditions, documented failure patterns, and helped validate fixes.
- Tuned thresholds, dwell-time rules, and trigger logic through iterative testing, implementing **10+ reliability improvements** across the workflow.

Software Engineer Intern, VMC

Mar 2024 – May 2024

- Used C# and .NET to analyze workflow defects, validate fixes, and document rollout steps for an online ticketing platform.
- Reproduced user-facing issues, reviewed system behavior, and communicated findings clearly for implementation and QA follow-up.

Projects

Credit Card Fraud Detection Using Transformer

Python, PyTorch, scikit-learn

- Built a **Transformer-based fraud detection model** on 284,807 transactions with 0.17% fraud prevalence, achieving **ROC-AUC 0.9643**, **Recall 0.8667**, and **Accuracy 0.9929**.
- Improved performance with a **Feature-Gated FT-Transformer** using Conv1D and sigmoid token gating, boosting **ROC-AUC to 0.9827** and reducing test loss from **0.3360 to 0.2758**.

CryptexLLM

Python, PyTorch, Transformers

- Built an **LLM-powered time-series forecasting pipeline** for BTC using OHLCV market data, patch-based tokenization, RevIN normalization, and a frozen LLM reprogramming head.
- Validated forecasting behavior with walk-forward evaluation, tracking MAE, RMSE, directional accuracy, trend behavior, and risk-adjusted returns under realistic fee assumptions.

E-Motel

React, Node.js, Express, MongoDB, Socket.IO, AWS, Vercel

- Built a **full-stack motel management web app** using MongoDB Atlas, Express, React, Node.js, JWT auth, and Tailwind CSS for room status, booking, customer, and staff workflows.
- Implemented **real-time updates with Socket.IO** and deployed the production build on Vercel/AWS, improving operational visibility and workflow coordination.

Real-Estate Management

React, Mapbox GL, GraphQL

- Built an interactive parcel management app using React, Mapbox GL, and GraphQL to support property search, parcel visualization, filtering, and spatial decision-making workflows.

Hackathons & Awards

MACIA Hackathon – Winner

Blockchain, Payment Processing

- Built and presented a **blockchain-based payment processing prototype** for secure transaction handling and transparent payment verification, winning the hackathon.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C#, C/C++

ML & Data: PyTorch, TensorFlow, scikit-learn, XGBoost, pandas, NumPy, matplotlib, seaborn, transformers, Hugging Face, model evaluation, classification, regression, Hadoop, Spark, Logistic Regression, Random Forest, Gradient Boosting, error analysis

Generative AI: OpenAI API, Hugging Face Transformers, LLM evaluation, prompt engineering, AI-assisted debugging, text generation and summarization concepts

Software & Web: React, Node.js, Express.js, MongoDB, REST APIs, GraphQL, Socket.IO, HTML/CSS, Tailwind CSS

Tools & Workflow: Git, Docker, AWS, Vercel, Linux, VS Code, Postman, code reviews, technical documentation, automation scripting

Professional: Analytical thinking, communication, collaboration, attention to detail, organization, reliable execution

Publication

Het Patel, et al. *CryptexLLM*

Under preparation

Research manuscript in preparation based on an LLM-powered time-series forecasting pipeline using OHLCV market data, RevIN normalization, and robustness evaluation across multiple training windows.